

Nrupendra Sai

Data Engineer || Data Analyst || Business Analyst || Big Data Engineer

Email: nrupendrasai@hotmail.com | Phone: (774) 459-6798

Professional Summary:

I am a data professional with 5+ years of experience in the field of data engineering and analytics. I am a dedicated and results-oriented engineer with experience in transforming complex datasets into actionable insights. I have a proven track record of leveraging statistical and analytical methods to uncover trends and patterns, providing valuable insights for informed decision-making. I am adept at designing and implementing data visualization tools to communicate findings effectively to both technical and non-technical stakeholders. I am proficient in SQL, Python, and data manipulation tools, with a keen attention to detail and a commitment to data accuracy. Experienced in collaborating cross-functionally to gather business requirements and translating them into meaningful data solutions. I am eager to leverage a strong foundation in statistical analysis and data interpretation to make a significant impact on data-driven decision-making processes.

- Designed and implemented data pipelines using Apache Spark and Apache Hadoop to process large datasets in a distributed manner.
- Developed and maintained data warehouses on Amazon Redshift and Snowflake to support data analytics and reporting.
- Created and optimized SQL queries to extract, transform, and load (ETL) data from various sources into data warehouses.
- Worked with cross-functional teams to collect and analyze business requirements to identify data-driven solutions.
- Presented data insights and findings to stakeholders to inform decision-making and drive business growth.
- Proficient in relational databases, including PostgreSQL, MySQL, and Oracle Database.
- Experience with cloud computing platforms, such as Amazon Web Services (AWS), Azure, GCP.
- Strong understanding of data modeling, data quality, and data security principles.
- Designed and implemented data pipelines using Python and SQL to efficiently move data between relational and NoSQL databases.
- Developed and maintained data models for relational and NoSQL databases to optimize data storage and retrieval.
- Utilized Tableau to create interactive data visualizations that effectively communicated key business insights to stakeholders.
- Implemented data quality checks and data governance procedures to ensure data accuracy and compliance with industry standards.
- Worked closely with data scientists to analyze large datasets and identify opportunities for data-driven decision-making.
- Presented data-driven recommendations to senior management to improve business processes and enhance customer experience.
- Experienced with NoSQL databases, such as MongoDB, Cassandra, and DynamoDB.
- Proficient in using SQL for data extraction, manipulation, and analysis, facilitating comprehensive and efficient database querying.
- Utilized Python and R for data analysis and scripting, automating repetitive tasks and streamlining analytical workflows.
- Applied data wrangling techniques to clean and preprocess raw data, ensuring data quality and accuracy for analysis.
- Conducted exploratory data analysis (EDA) to identify patterns, outliers, and trends, informing strategic decision-making processes.
- Utilized advanced Excel functions and features for data analysis, including pivot tables, VLOOKUP, and statistical functions.
- Collaborated with data engineers to understand and optimize data structures, ensuring data availability and accuracy for analysis.
- Integrated and analyzed data from various sources, including APIs and external databases, to provide comprehensive insights.
- Developed and maintained interactive dashboards for real-time monitoring of key performance indicators (KPIs) using tools like Grafana.
- Implemented geospatial analysis techniques to uncover location-based insights and trends within datasets.
- Implemented data profiling techniques to assess data quality and identify potential data issues.
- Utilized version control systems, such as Git, for tracking changes in analytical code and ensuring reproducibility.

- Collaborated with IT professionals to streamline data integration processes, ensuring seamless access to diverse datasets.
- Provided analytical support to sales teams, aiding in the identification of growth opportunities and customer segmentation strategies.
- Collaborated with operations teams to streamline data collection processes, reducing data latency and improving real-time reporting.
- Engaged in cross-functional workshops to define data governance policies and ensure data quality standards.

Technical Skills:

Methodologies	: Scrum and Agile
Programming Languages	: Python, SQL, R, Spark, Java Script, MySQL, SnowflakeSQL
Frameworks	: Django, Flask, Rails, Sinatra
IDE'S	: Jupyter Notebook, PyCharm, Visual Studio, IntelliJ
Development Tools	: AirFlow, Docker, Kubernetes, Amazon S3, Excel
Databases	: MySql, Oracle, MongoDB, PostgreSQL, Microsoft SQL, Snowflake
Data Warehousing	: Spark, Hadoop, Amazon Redshift, Apache Kafka, Apache Flink
Version Controls	: Git, GitHub
Cloud	: AWS (Amazon Web Services), Azure, GCP
Data Visualization	: Power BI, Tableau, Excel
Operating Systems	: Linux, Windows, MacOS

Professional Experience:

US Courts - Baltimore, Maryland

Jan 2023 - Current

Data Engineer|Data Analyst (Airflow|Power BI|Azure)

- Implemented the design and implementation of DataForge's ETL pipelines, harmonizing diverse data sources and achieving a 40% reduction in data processing time.
- Conducted the migration to Azure, creating a robust cloud data lake that facilitated seamless scalability and storage for terabytes of mission-critical data.
- Implemented Apache Flink for distributed data processing, enhancing DataForge's ability to process real-time streaming data efficiently.
- Engineered a real-time analytics engine using Apache Kafka and Flink, enabling DataForge to gain immediate insights from streaming data sources.
- Crafted intricate data models using a combination of ER diagrams and dimensional modeling, ensuring optimal storage efficiency and ease of retrieval.
- Conducted in-depth performance tuning on MongoDB databases, achieving a 25% improvement in query response times for high-velocity data.
- Leveraged Python scripting for automating complex data transformations, significantly reducing manual intervention and minimizing errors.
- Instituted robust security measures in SQL Server databases, ensuring compliance with industry regulations and safeguarding sensitive information.
- Implemented Git for version control, providing a centralized repository for collaboration and ensuring traceability of changes across the DataForge project.
- Established and enforced data governance policies and ensured data quality and integrity.
- Designed and implemented a data warehousing solution using Amazon Redshift, optimizing storage and retrieval for analytical processing.
- Collaborated seamlessly with data scientists, analysts, and business stakeholders, ensuring alignment between data solutions and organizational objectives.
- Conducted Exploratory Data Analysis and visualized data with Tableau/PowerBI to present analytical findings and insights to stakeholders and management.

- Developed Data Warehousing and Business Intelligence solutions using Azure Functions, Azure Data Factory, SQL Server, NoSQL for content providers.
- Instituted a robust CI/CD pipeline using Jenkins, automating testing and deployment processes for DataForge's data engineering workflows.
- Involved in writing Spark jobs using PySpark.
- Strong experience in Business and Data analysis, Data Profiling, Data Migration, Data Integration and Metadata Management Services
- Strong experience in Data Modeling with expertise in creating Star and Snow-Flake Schemas, FACT and Dimension Tables, Physical and Logical Data Modeling using Erwin.
- Experience in understanding Stored Procedures, Stored Functions, database triggers, packages using PL/SQL.
- Designed and maintained robust ETL pipelines, ensuring efficient and reliable data processing for analytical purposes.
- Implemented Docker for containerization of Data Forge applications, ensuring consistency and reproducibility across various environments.
- Applied geospatial analysis techniques to uncover location-based insights within datasets, providing valuable spatial context to Data Forge's analytics.
- Designed and developed dynamic dashboards using Tableau, providing stakeholders with real-time insights into key performance indicators.
- Implemented data anonymization and pseudonymization techniques, ensuring compliance with stringent data privacy regulations.
- Pioneered data quality initiatives, implementing automated data validation checks and ensuring consistency and accuracy in DataForge's data processing.
- Facilitated knowledge-sharing sessions, enhancing cross-functional collaboration and understanding of data processes at DataForge.

Kate Technologies Pvt Ltd - Hyderabad, India

Aug 2019 - Jan 2022

Data Engineer|Data Analyst| Big Data Engineer (Tableau|AWS|Hadoop|)

- Leveraged AWS Glue and Amazon Data Pipeline to develop and implement serverless data integration pipelines, seamlessly integrating data from production databases (SAP S4HANA), sensor data, and external datasets.
- Enhanced data quality by implementing data cleansing and validation procedures using AWS Glue's built-in functions and Python libraries like pandas and NumPy, ensuring the accuracy and reliability of data used for analysis.
- Streamlined data collection processes by automating data extraction and ingestion from various sources, including SAP S4HANA, Oracle EBS, and Salesforce, using AWS Lambda and Amazon Kinesis, reducing manual intervention and improving data availability.
- Uncovered hidden patterns and trends within complex manufacturing and engineering datasets using Amazon Redshift's SQL capabilities and Python libraries like pandas, matplotlib, and seaborn, employing exploratory data analysis techniques.
- Employed statistical analysis and hypothesis testing using R and Python libraries like scipy and statsmodels, validating data-driven insights and providing a rigorous foundation for strategic decision-making.
- Identified anomalies and outliers in production and quality data using Tableau and Power BI, enabling timely intervention and corrective actions to prevent process disruptions.
- Communicated complex data insights effectively to both technical and non-technical stakeholders using Tableau and Power BI, creating interactive dashboards and reports that provided real-time visibility into manufacturing performance, quality metrics, and operational efficiency.
- Developed data storytelling techniques using Tableau and Power BI, engaging stakeholders and driving data-driven decision-making across the organization.
- Created visually appealing and informative dashboards and reports using Tableau and Power BI, effectively summarizing key metrics and trends for easy comprehension.

- Utilized machine learning algorithms implemented in Python libraries like scikit-learn and TensorFlow to develop predictive models that forecast manufacturing output, product quality, and operational outcomes, enabling data-driven planning and decision-making.
- Enabled data-driven decision-making by providing insights that guided production planning, quality control initiatives, and resource allocation, optimizing resource utilization and enhancing overall efficiency.
- Investigated production issues and quality concerns by analyzing historical and real-time data stored in Amazon Redshift using SQL queries and Python libraries like pandas and matplotlib, identifying root causes and contributing factors.
- Developed data-driven recommendations for corrective actions, utilizing Python and Tableau to present findings and insights to stakeholders, leading to process improvements and reduced downtime.
- Implemented data-driven solutions to prevent recurring issues, enhancing manufacturing efficiency and product quality, by incorporating data-driven insights into production processes and quality control procedures.
- Software Development Life Cycle (SDLC) experience including Requirements, Specifications Analysis/Design and Testing as per the SDLC.
- Designed and Implemented data integration modules for Extract/Transform/Load (ETL) functions.
- Extensively involved in Data Extraction, Transformation and Loading (ETL process) from Source to target systems using informatica.
- Established and maintained key performance indicators (KPIs) using Tableau and Power BI, tracking manufacturing and engineering metrics stored in Amazon Redshift.
- Identified areas for improvement and optimization based on data-driven insights, driving continuous improvement initiatives.
- Collaborated closely with engineering, manufacturing, and quality assurance teams to understand their data needs and provide analytical support, utilizing AWS tools like Amazon SageMaker and Amazon Comprehend to build and deploy machine learning models for predictive analytics.
- Presented data findings and recommendations to cross-functional teams, fostering a data-driven culture and enabling data-informed decision-making, leveraging AWS services like Amazon QuickSight and Amazon Athena for data exploration and visualization.
- Shared knowledge and expertise with colleagues, promoting data literacy and data-driven problem-solving across the organization, utilizing AWS Glue and Amazon Data Pipeline to democratize data access and enable self-service analytics.
- Successfully delivered multiple SDLC projects using AGILE methodology, from requirements gathering to architecture definition, design, development, modeling, and testing.
- Analyzed and fetched data from complex databases using SQL statements, joins, and procedures.
- Used Git for version control and created automated build scripts.
- Mentored and trained junior data analysts, fostering a culture of data-driven decision-making within the organization.

Kate Technologies Pvt Ltd - Hyderabad, India

Jul 2017 - Jul 2019

Business intelligence analyst Data Analyst (Tableau|Excel| VLOOKUP)

- Harvested actionable insights from raw data utilizing a combination of Tableau, Excel, VLOOKUP, and dashboarding techniques.
- Formulated and implemented data-driven recommendations to optimize business operations, leveraging Tableau for data visualization and Excel for data manipulation.
- Collaborated closely with internal and external clients to identify and address analytical requirements, employing Tableau to communicate insights effectively.
- Generated ad hoc data and reports for upper management decision-making, utilizing Excel and VLOOKUP to extract and summarize key metrics.
- Unveiled hidden patterns and trends within complex datasets by applying advanced data analysis techniques, including Tableau's interactive data visualizations.

- Constructed predictive models using Tableau and Excel to forecast future business outcomes and inform strategic planning initiatives.
- Persuasively presented data-driven findings to senior management, influencing strategic decision-making and resource allocation through Tableau's storytelling capabilities.
- Developed and maintained a comprehensive data repository using Tableau's data governance features to ensure data accuracy and accessibility.
- Adhered strictly to data governance principles to safeguard sensitive information and maintain data integrity, utilizing Excel's data protection tools.
- Demonstrated exceptional problem-solving skills, effectively identifying and resolving data anomalies and inconsistencies through Tableau's data cleaning and validation functionalities.
- Initiated a culture of data-driven decision-making, fostering a data-informed approach across all levels of the organization by promoting the adoption of Tableau and Excel as analytical tools.
- Designed and built dimensional databases to meet business needs, ensuring data integrity and accessibility.
- Managed, analyzed, and resolved data initiative issues, adhering to published data standards and best practices.
- Assisted in implementing data standards and guidelines on data ownership, coding structures, and data replication.
- Collaborated with business process experts to define standard operating procedures, process flows, and related documentation.
- Conducted data cleaning to rid systems of old, unused, or duplicate data, improving management and access.
- Developed and implemented strategies to translate business requirements into feasible data warehouse designs.
- Provided data consulting to clients, improving their database systems and supporting business initiatives.
- Created and maintained Master Records such as Material, Vendor, and Purchase Info records in Tableau.
- Conducted the design and implementation of a data warehouse solution that significantly improved data accessibility and reduced query time by 20%.
- Played a pivotal role in cleansing and standardizing massive datasets, ensuring data integrity and enabling accurate analysis.
- Developed data quality metrics and dashboards to monitor data quality, identify anomalies, and track progress on data quality improvement initiatives.
- Presented data findings and recommendations to senior management, influencing strategic decision-making.
- Adhered to strict data governance policies and procedures, ensuring data security and compliance with regulatory requirements.

Education:

Lamar University - Beaumont, Texas
Master of Science in Computer Science

Jan 2022 - Aug 2023

Manipal Institute of Technology - Manipal, India
Bachelor of Technology in Electrical and Electronics Engineering

Aug 2013 - Jun 2017

Projects:

Disaster Response Pipeline - Python, Natural Language Processing, Flask, sqlite, pandas

- Developed a web application utilizing NLP techniques to classify emergency messages into distinct categories.
- Implemented an ETL pipeline facilitating efficient data reading, cleaning and storage in an SQLite database.
- Built a machine learning pipeline using Random Forest to classify 36 distinct categories.
- Deployed a Flask-based web-app that utilizes a pre-trained model for display.
- Predicted categories based on user-input text messages with 95% accuracy.

Image based PM2.5 Pollution Analysis - Matlab, Autoencoder, Image Processing

- Developed a lightweight image-based method for PM2.5 concentration estimation.
- Utilized a 3-layer autoencoder with saturation and super-pixel techniques for image feature extraction.
- Model validation on a 1460-photo dataset achieving results comparable to state-of-the-art methods (SoftMax & Gaussian regression).